STA 431 / MIDTERM MENU CARD

TI-84 Plus CE | Freund Chapters 1-3 | Homework 1-3

START PRGM > STA431 CALC = CALCULATORS | ENC = ENCYCLOPEDIA | RV = RANDOM VARIABLES | (-)1 = unknown

HOMEWORK 1 | COUNTING

Q1 Pascal identity + story proof
ENC > CH1 > BINOMIAL COEFFS > PASCAL IDENTITY
C(n,r) = C(n-1,r-1)+C(n-1,r). Algebra: common denominator. Story: fix one item.

Q2 Two binomial-sum proofs
ENC > CH1 > BINOMIAL COEFFS > SUM IDENTITIES
Differentiate (1+x)^n at x=1; integrate it from 0 to 1. Results: n·2^(n-1); (2^(n+1)-1)/(n+1).

Q3 $x^3y^2z^2$ term coefficient
CALC > COUNT > 6
Enter (2,3), (-1,2), (3,2). Rule: 7!/(3!2!2!) × $2^3(-1)^2 3^2$.

Q4 PROBABILITY arrangements
CALC > COUNT > 2
Enter word; choose LETTER BLOCK B for adjacent B's. Total 11!/(2!2!); block 10!/2!.

Q5 Three distinct committees
CALC > COUNT > 3 > 1
Sizes 3,4,5; LABELED=1. Rule: 12!/(3!4!5!).

Q6 T/F exam: 15 right, 2 hard
CALC > COUNT > 4
A=5 hard, B=15 other, K=15; hard min=max 2. C(5,2)C(15,13).

Q7 Delegation: at least 2 each
CALC > COUNT > 4
A=8 math, B=6 stats, K=5; min 2 each. Sum C(8,i)C(6,5-i), i=2,3.

Q8 Grades: exact A and F
CALC > COUNT > 3 > 2
N=10; exact counts 3 A and 1 F; remaining 6 choose among B,C,D (3).

HOMEWORK 2 | PROBABILITY

Q1 Translate event statements
ENC > CH2 > SETS & EVENTS > ENGLISH ► SETS
Only A = A∩B'∩C'; exactly one = union of the three only-regions.

Q2 Union + conditionals
CALC > PROB > 1
A=.55, B=.35, union=.70; unknowns=(-)1. Intersection=A+B-union; P(A|B)=intersection/B.

Q3 Graduate/faculty 2×2
CALC > PROB > 2
Rows grad/faculty, cols female/male. Enter grad-female=40, row totals 70/50, female=65, N=120; rest=(-)1.

Q4 Journal acceptance
CALC > PROB > 3
Priors .5,.3,.2; accept rates .15,.25,.10. Bayes posterior = prior×rate / total.

Q5 Die selects an urn
CALC > PROB > 3
Cases A/B: priors 1/3,2/3; red likelihoods 3/7,5/7. Use total probability + Bayes.

Q6 Spam + urgent
CALC > PROB > 3
Cases spam/not: priors .6,.4; urgent likelihoods .4,.05.

Q7 Weekend only / none
CALC > PROB > 5 > 3
A=Digital, B=Print, C=Weekend. Enter pairs, triple, N=1000. C-only=C-AC-BC+ABC; none=N-(A+B+C-AB-AC-BC+ABC).

HOMEWORK 3 | RANDOM VARIABLES

Q1 Maximum of two balls
CALC > RV > 2 > MAXIMUM > WITHOUT
N=5, k=2. p(z)=(z-1)/C(5,2); F(z)=C(z,2)/C(5,2). Sketch spikes + right-continuous steps.

Q2 Two-interval PDF to CDF
CALC > RV > 3 > PDF
Two pieces: (0,2) and (4,6), each constant 1/4. Integrate each; CDF stays flat on (2,4).

Q3 Given CDF: interval + tail
CALC > RV > 3 > CDF
One piece (-2,2), F(x)=1/2+x/4. Use F(b)-F(a) and 1-F(a).

Q4 Train polynomial PDF
CALC > RV > 3 > PDF
One piece (-4,4): c0=3/16, c2=-3/256. Integrate intervals; P(X=-1)=0.

Q5 Normalize cy^2, then CDF
CALC > RV > 3 > PDF
Piece (0,3): UNKNOWN c=YES, c2=1. c=1/9, F(y)=$y^3/27$; tail=1-F(2).

Q6 Exponential CDF to PDF
ENC > CH3 > CONTINUOUS RVs > PDF ◄► CDF; CALC > RV > 4
F=1-e^(-2x), x>0. Symbolically f=F'=2e^(-2x); tail P(X>1)=e^(-2). Use [0,50] for numeric check.

FAST RULES Bayes = prior × likelihood / evidence | continuous P(a<X<b)=F(b)-F(a), f=F', P(X=a)=0

CH3 JOINT CALC > RV > 6 JOINT X,Y: marginals, conditionals, independence. Show proofs and sketches by hand.